

**ADDIS ABABA UNIVERSITY
DEPARTMENT OF CHEMISTRY**

**FUNDAMENTALS OF ORGANIC CHEMISTRY
COURSE OUTLINE**

1. Introduction

- 1.1. Historical background of organic chemistry
- 1.2. Atomic structure of carbon, covalent bond and hybridization
- 1.3. Molecular, structural and three-dimensional formula of organic compounds
- 1.4. Lewis Structure and Resonance
- 1.5. Factors influencing electron availability and reactivity of organic compounds
Inductive effect, Resonance effect and Steric effect
- 1.6. Functionality and classes of organic compounds

2. Nomenclature of organic compounds

Alkanes, Alkenes, Alkynes, Alcohols, Aromatics, Aldehydes, Ketones, Amines, Ethers, carboxylic acids, carboxylic acid derivatives (Esters, Amides, acid anhydrides, acyl halides)

3. Stereochemistry

- 3.1. Stereoisomerism: Definition and Classes (Geometric, Configurational, Conformational Isomerism)
- 3.2. Definitions: Superstability, chirality
- 3.3. Test for Chirality (plane of symmetry)
- 3.4. Enantiomers and their Properties
- 3.5. Configuration of Chiral Compounds (The Cahn-Ingold-Prelog (CIP) sequence rules for assigning configurations)
- 3.6. Optical Activity (Measurement of Optical Rotation: The Polarimeter)
- 3.7. Fischer Projections
- 3.8. Multiple Stereogenic Centers: Diastereomers
- 3.9. Meso compounds
- 3.10. Racemic mixtures and their properties
- 3.11. Biological importance of Chirality
- 3.12. Conformational Isomerism
 - 3.12.1. Conformational Analysis in alkanes: Ethane and n-Butane
 - 3.12.2. Cycloalkanes: Cyclopropane, cyclobutane, cyclopentane and Cyclohexane
 - 3.12.3. Substituted Cycloalkanes: mono, di and trisubstituted cyclohexanes

4. Major Organic Reactions

- 4.1. Substitution reactions
 - 4.1.1. Introduction (Curved arrows, Nucleophiles, Electrophiles, definition and examples)
 - 4.1.2. S_N1 and S_N2 mechanisms
 - 4.1.3. Factors affecting S_N2 and S_N1 reactions
 - 4.1.4. Applications of substitution Reactions
Alkyl halides (Williamson ether synthesis, Preparation of primary amine, preparation of Alkynes), preparation of alkyl halides from alcohols, Ethers, Epoxide Ring opening
Elimination reactions
- 4.2. Elimination reactions
 - 4.2.1. E1 and E2 Mechanisms
 - 4.2.2. Elimination versus substitution
 - 4.2.3. Applications of elimination reactions
Dehydration of Alcohols, Dehydrohalogenation of alkylhalides, Vicinal Dihalides, Hofmann Elimination, Acetate pyrolysis, Cope reaction
- 4.3. Addition Reactions
 - 4.3.1. Rules of addition reactions
Markovnikov Rule, Michael Addition, Radical addition

- 4.3.2. Applications of Addition Reactions
Hydrogenation, Halogenation, Addition of hydrogen halide, Addition of water, Hydroboration, Halohydrine formation, ozonolysis, Hydroxylation.
- 5. Aromatic Chemistry**
 - 5.1. Aromaticity
 - 5.2. Electrophilic aromatic substitution reaction (EAS)
 - 5.3. Directing effects of substituents on EAS
 - 5.4. Application of electrophilic substitutions
Halogenation, Sulfonation, Nitration, Friedel-Crafts Alkylation, Friedel-Crafts Acylation, Diazotization of Amines, Reactions of aromatic side chains
- 6. Carbonyl reactions**
 - 6.1. Carbonyl addition
 - 6.2. Addition Elimination
 - 6.3. Enolization-Ketonization
 - 6.4. Application of Addition reactions
Hydrate formation, Hemiacetals and Hemiketals, Cyanohydrins, Carbinolamines, Addition of Grignard reagents, Addition of hydrogen, Lithiumaluminiumhydride and sodiumborohydride.
 - 6.5. Applications of addition-elimination reactions
Imines and related compounds, Wittig reaction, Acetal formation in acid media, Acids and their derivatives, Ester hydrolysis and formation in acid media, Acid chlorides, Acid anhydrides, Reduction of acid derivatives
 - 6.6. Application of enolization-Ketonization reactions
Halogenation, Alkylation, Aldol condensation, Claisen-Schmidt condensation, Mannich condensation, Perkin condensation, Claisen condensation
- 7. Oxidation-reduction reaction**
 - 7.1. Oxidation Reaction
Alcohols and Aldehydes
 - 7.2. Reduction reactions
Catalytic Hydrogenation, Chemical Reduction, Dissolving metal reductions
- 8. Biological molecules**
 - 8.1. An introduction to carbohydrate chemistry
Monosaccharides (classification, cyclic structure, epimers)
Reaction of monosaccharides (Ester and ether formation, glycoside formation, Reduction, Oxidation)
 - 8.2. Disaccharides and polysaccharides
 - 8.3. Amino-acids, peptides, and proteins
 - 8.3.1. The structure and properties of alpha-amino acids
 - 8.3.2. Analysis of alpha-amino acids
 - 8.3.3. Synthesis of alpha-amino acids
 - 8.4. Peptides and proteins
 - 8.5. Peptide synthesis
 - 8.6. Introduction to Chemistry of Lipids
 - 8.7. Introduction to Chemistry of Nucleic Acids

REFERENCE

1. T.W G. Solomons, Organic Chemistry, 7th ed., 2004.
2. J. Mc Murry, Organic Chemistry, 7th ed., 2008.
3. F. A. Carey, Organic Chemistry, 3rd ed., 1996.